Introduction to AI for Cybersecurity

Today's goal • 3/3 items

Module 1
Course Introduction

Module 2
AI for Cybersecurity Professionals

Module 3
Ham or Spam? Detecting Email Cybersecurity Threats with AI

AI Techniques for Detecting Email Threats

- Detecting Email Cyber Threats with AI
  Video • 4 min
- Perceptron Spam Filter Example
  Video • 2 min
- Support Vector Machine (SVM) Spam Filter Example
  Video • 3 min
- Reading References
  Reading • 15 min
- AI Techniques for Detecting Email Threats
  Practice Assignment • 15 min

Advanced Spam Filtering Methods and Phishing Detection

- Phishing Detection with Regression and Trees
  Video • 3 min
- Naive Bayes Spam Filter Example
  Video • 1 min
- Reading References
  Reading • 15 min
- Advanced Spam Filtering Methods and Phishing Detection
  Practice Assignment • Grade: 100%

Module-end Assessments

- Self-Reflective Reading: Bridging Theory and Practice in Cybersecurity
  Reading • 40 min
- Ham or Spam? Detecting Email Cybersecurity Threats with AI
  Graded Assignment • 1h

Module 4
Securing User Authentication

Self-Reflective Reading: Bridging Theory and Practice in Cybersecurity

0:11 / 3:23 2x

Introduction

As cybersecurity threats evolve, so do the tools designed to combat them. One of the key areas where Artificial Intelligence (AI) is making an impact is spam filtering. Modern AI-powered spam filters can detect and block malicious emails with increasing accuracy. However, how do these newer AI approaches compare to traditional methods like the NLP-Naïve Bayes classifier? This module encourages learners to not only review the current state of AI spam filters but also reflect on their implications for cybersecurity and their professional practice.

This task will also prompt learners to consider the practical benefits of accurate spam filters in the work of a cyber analyst, focusing on how AI can streamline their responsibilities. Lastly, by exploring the development of these tools through the lens of machine learning (ML), learners will gain deeper insights into how AI systems are created and improved.

Task Overview

1. **Literature Review**: Conduct a literature review on the most recent AI spam filters.
2. **Comparison**: Reflect on and describe the differences between the current AI spam filters and the NLP-Naïve Bayes classifier.
3. **Cybersecurity Application**: Consider how an accurate AI spam filter could improve a cyber analyst's daily workflow and decision-making.
4. **ML Development Process**: Reflect on the steps involved in developing AI spam filters, and summarize the process using the ML analytic framework.

Advised Steps

1. Research and Reflect:

- Start by gathering recent literature on AI spam filters.
- As you review the material, think about what each approach offers and how it improves upon older methods.

2. Analyze and Compare:

- Compare the strengths and limitations of modern AI spam filters with those of the NLP-Naïve Bayes classifier.
- Reflect on which aspects are most critical for today's cybersecurity challenges.

3. Consider Practical Impact:

- Think about the role of a cyber analyst and how the accuracy of spam filters can make their work easier.
- What specific tasks could be automated or made more efficient? Reflect on the potential for reducing errors and increasing security.

4. Use the ML Framework:

- Reflect on the stages of the ML development process (such as data collection, feature selection, model training, and evaluation) and how these apply to the AI spam filters you reviewed.
- Consider how these steps contribute to building a more effective spam filter.

Self-Reflective Questions:

- During my literature review, what differences between modern AI spam filters and the NLP-Naïve Bayes classifier stood out to me? Why are these differences important for the future of cybersecurity?
- How do I believe accurate AI spam filters could change the role of a cyber analyst? In what ways could they improve workflow, reduce manual tasks, or enhance decision-making?
- What new insights did I gain from understanding the ML development process in the context of spam filters? How does this process influence the quality and accuracy of the tools?
- How might I communicate the benefits of AI spam filters and the importance of their development to my colleagues or within my organization?
- Reflecting on what I have learned, how can I apply this knowledge to improve cybersecurity practices in my professional setting?

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →